Path Integral Derivation of the Aharonov-Bohm Effect

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Abstract

In this note, we derive the Aharonov-Bohm effect using the path-integral approach to quantum mechanics. In particular, we derive the probability-amplitude distribution on the detection screen for a double-slit-like experiment with a solenoid between source and detection screen. We use a heuristic argument to truncate a sum over homotopy classes in the propagator to derive the interference pattern observed at the detection screen due to the magnetic flux Φ through the solenoid.

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1 Introduction

The Aharonov-Bohm effect [1] demonstrates the fundamental importance of potentials in quantum mechanics. Potentials in, say electromagnetism, are defined for mathematical utility, but in quantum mechanics, they are of fundamental physical significance as the Aharono-Bohm effect demonstrates.

In this note, we show, using the path integral approach to quantum mechanics [3, 9, 6] provides a natural method of investigating this effect by using classical intuition to derive a quantum effect. We discuss this derivation, which can also be found in [7, 2, 8, 4] to varying levels of explicitness. We attempt to make this derivation mathematically explicit, using elements of homotopy theory B.

2 Path Integrals

The quantum amplitude to find a quantum particle at the point x_f at time t_f is given by the wavefunction $\Psi(x_f, t_f) = \langle x_f|\Psi(t_f)\rangle$ which can be written as an integral

$$\Psi(x_f, t_f) = \int dx_i K(x_f, t_f : x_i, t_i)\Psi(x_i, t_i).$$

The causal Green's function $K = K(x_F, t_f : x_i, t_i)$ is called the **propagator** and can be, in principle, computed if the action of the Hamiltonian on the position basis is known,

$$K = \langle x_f| U(t_f, t_i) |x_i\rangle .$$

There is an alternative way to compute K, which, instead of knowing the Hamiltonian of the system, requires the knowledge of only the *classical* Lagrangian of the corresponding classical system. This formalism is called the **path integral formulation** of quantum mechanics, and is the statement that K is a sum over all possible histories of the system - with appropriate boundary conditions - of global phases dependent on the *classical* action of the system. For the case of a single particle, the histories - with appropriate boundary conditions - of the system are simply the paths of the particle through physical (or configuration) space from the spacetime point (x_i, t_i) to the spacetime point (x_f, t_f),

$$K = \sum_{\text{paths}} e^{\frac{i}{\hbar}S[\text{path}]}.$$

More generally, for a system described by generalized coordinates and momenta on a phase space, (q, p), and with Hamiltonian of the form $H = T(p,t)+V(x,t)$, the propagator is a phase-space path integral over histories in phase-space.

This formulation of quantum mechanics seems like it would be easier to compute things with, since we only deal with classical quantities. However, the main difficulty with the path integral formulation is that defining the "sum over histories" for particular systems and using it to actually compute an integral can be extremely complicated mathematically, so that you are no longer doing physics but rather mathematics. Nonetheless, there are many systems where the "sum over histories" is very well defined, and even for those for which it is not, path integral methods offer the advantage of physical intuition, because of the classical quantities appearing in the formulation. In these notes, we derive the "sum over histories" of a single particle in a time independent Hamiltonian of the form $H = T + V$ - so that the Lagrangian, being the Legendre transform of the Hamiltonian, is of the form $L = T - V$ - and use the path integral formulation to demonstrate the renown **Aharonov-Bohm effect**.

In general, for a system described by generalized coordinates q^i and generalized momenta p_i, the propagator is

$$K = \int \left(\prod_i \mathcal{D}q^i \mathcal{D}p_i \right) e^{\frac{i}{\hbar}S[\mathbf{q},\mathbf{p}]},$$

or, for a field theory described by the fields ϕ^i and their conjugate momenta π_i,

$$K = \int \left(\prod_i \mathcal{D}\phi^i \mathcal{D}\pi_i \right) e^{\frac{i}{\hbar}S[\phi,\boldsymbol{\pi}]}.$$

3 Aharonov-Bohm

Consider a point particle with mass m and charge q moving in 2-dimensions. Furthermore, consider an infinitely long solenoid perpendicular to the plane of motion of the particle so that there is only a magnetic field $\mathbf{B} = \nabla \times \mathbf{A}$ inside the solenoid, along its axis of cylyndrical symmetry. The inside of the solenoid is an inaccessible region for the point particle and therefore never comes into contact locally with the magnetic field. However, the magnetic vector potential $\mathbf{A}(\mathbf{x})$ need not vanish outside of the solenoid.

An infinitely long solenoid of radius R aligned along the z-axis has magnetic field parallel to the z-axis inside ($r < R$) the solenoid, $\mathbf{B}(\mathbf{r}) = B\hat{\mathbf{z}}$, and zero magnetic field outside ($r > R$) the solenoid. Therefore, any valid vector potential must satisfy $\nabla \times \mathbf{A}(\mathbf{x}) = B\hat{\mathbf{z}}$ inside of the solenoid and $\nabla \times \mathbf{A} = 0$ outside of the solenoid. In general, even though the magnetic field is zero outside of the solenoid, the vector potential does not need to be zero.

The Lagrangian for a non-relativistic particle with mas m and charge q in a vector potential field is

$$L = \frac{1}{2}m\dot{\mathbf{x}}^2 - V(\mathbf{x}) + \frac{q}{c}\dot{\mathbf{x}} \cdot \mathbf{A}(\mathbf{x}).$$

Therefore, the action for the path γ of a particle through the space $\mathcal{M} = \mathbb{R}^2 - \Sigma$ is

$$S[\gamma] = \int_{t_i}^{t_f} dt \left(\frac{1}{2}m\mathbf{x}^2 - V(\mathbf{x})\right)\Bigg|_\gamma + \frac{q}{c}\int_{t_i}^{t_f} dt\, \dot{\mathbf{x}}\cdot\mathbf{A}(\mathbf{x})\Bigg|_\gamma .$$

The paths $\gamma : [0,1] \to \mathcal{M}$ are understood to be paths from the source at $\mathbf{x} = \gamma(0) = \mathbf{x}_i$ to the detection screen at $\mathbf{x} = \gamma(1) = \mathbf{x}_f$. We denote the space of all paths from $\mathbf{x}_i$ to $\mathbf{x}_f$ by Γ. Since $\dot{\mathbf{x}} = \frac{d\mathbf{x}}{dt}$, we have that

$$S[\gamma] = S_0[\gamma] + \frac{q}{c}\int_\gamma d\mathbf{x}\cdot\mathbf{A}(\mathbf{x}),$$

where $S_0[\gamma]$ is the *mechanical action* (since it is the mechanical contribution, as opposed to the electromagnetic contribution), and the electromagnetic contribution only depends on the path, and not the time parameterization of the path, and thus also not on the velocity of the path. So, the propagator for the particle to go from $(\mathbf{x}_i, t_i)$ to $(\mathbf{x}_f, t_f)$ is

$$\begin{aligned} K(\mathbf{x}_f, t_f : \mathbf{x}_i, t_i) &= \int_\Gamma \mathcal{D}\gamma\, e^{\frac{i}{\hbar}S[\gamma]} \\ &= \int_\Gamma \mathcal{D}\gamma\, \exp\left[\frac{i}{\hbar}S_0\right]\exp\left[\frac{iq}{\hbar c}\int_\gamma d\mathbf{x}\cdot\mathbf{A}(\mathbf{x})\right]. \end{aligned}$$

Since the space $\mathcal{M}$ is topologically non-trivial - i.e., it has a region which is inaccessible to the particle, represented by a hole giving a non-trivial fundamental group $\pi_1(\mathcal{M}) \cong \mathbb{Z}$ - the space of all paths Γ can be partitioned into topologically distinct sectors by homotopy class $[\gamma_n]$. γ_n is a representative of the equivalence class of paths $[\gamma_n]$ which wind n times around Σ. The way we define the winding number of an open path from $\mathbf{x}_i$ to $\mathbf{x}_f$ is by choosing a reference path $\bar{\gamma}_0 \in [\gamma_0]$ which we define to have winding number zero, and by considering an arbitrary path $\bar{\gamma}_n$, from $\mathbf{x}_i$ to $\mathbf{x}_f$. If we let γ^{-1} denote the path γ parameterized in the opposite direction, the composition $\bar{\gamma}_0^{-1}\circ\bar{\gamma}_n$ is a loop based at $\mathbf{x}_i$ and thus has a known winding number, n. An example is shown in Fig. [?], with $n = 2$. Thus, we say that the path $\bar{\gamma}_n$ has winding number n iff $\bar{\gamma}^{-1}\circ\bar{\gamma}_n \in [\gamma_n]$ has winding number n. For example, and in particular (since we will use the following example in the double slit experiment), the loop associated with the path which winds around Σ zero times but from below, so that it cannot be continuously deformed into the reference path, is $\bar{\gamma}^{-1}\circ\bar{\gamma}_1$ which has winding number 1. Therefore, we say that $\bar{\gamma}_1$ has winding number 1 relative to $\bar{\gamma}$ so that it is in the winding number 1 equivalence class $[\gamma_1]$.

The propagator can then be split into a sum over each topological sector:

$$K(\mathbf{x}_f, t_f : \mathbf{x}_i, t_i) = \sum_{n\in\mathbb{Z}}\int_{\gamma_n\in[\gamma_n]} \mathcal{D}[\gamma_n]\, \exp\left[\frac{i}{\hbar}S_0[\gamma_n]\right]\exp\left[\frac{iq}{\hbar c}\int_{\gamma_n} d\mathbf{x}\cdot\mathbf{A}(\mathbf{x})\right].$$

Using this definition of winding number, we can evaluate the electromagnetic contribution to the action:

$$\oint_{\gamma^{-1}\circ\gamma_n} d\mathbf{x}\cdot\mathbf{A}(\mathbf{x}) = \int_{\gamma_n} d\mathbf{x}\cdot\mathbf{A}(\mathbf{x}) - \int_{\gamma_0} d\mathbf{x}\cdot\mathbf{A}(\mathbf{x}).$$

By Stoke's theorem,

$$\oint_{\gamma_0^{-1}\circ\gamma_n} d\mathbf{x}\cdot\mathbf{A}(\mathbf{x}) = n\iint_\Sigma d\mathbf{a}\cdot(\nabla\times\mathbf{A}(\mathbf{x})) = n\Phi. \tag{1}$$

Therefore,

$$\int_{\gamma_n} d\mathbf{x}\cdot\mathbf{A}(\mathbf{x}) = \int_{\gamma_0} d\mathbf{x}\cdot\mathbf{A}(\mathbf{x}) + n\Phi.$$

The propagator is then

$$K(\mathbf{x}_f, t_f : \mathbf{x}_i, t_i) = \exp\left[\frac{iq}{\hbar c}\int_{\bar{\gamma}_0} d\mathbf{x}\cdot\mathbf{A}\right]\sum_{n\in\mathbb{Z}} e^{i\frac{qn\Phi}{\hbar c}} K_n^{(0)}(\mathbf{x}_f, t_f : \mathbf{x}_i, t_i).$$

Now, we can truncate the sum to paths with winding numbers 0 and 1 since those terms contribute negligibly in the semi-classical approximation. From a semi-classical interpretation, paths with $n \neq 0, 1$ obviously have longer path lengths and thus are not the extremizing paths of the classical action. Therefore, the propagator can be written as approximately

$$K \approx \exp\left[\frac{iq}{\hbar c}\int_{\bar{\gamma}_0} d\mathbf{x}\cdot\mathbf{A}\right]\left(K_0^{(0)} + e^{i\delta}K_1^{(0)}\right). \tag{2}$$

Now, if the source emits localized charged particles (so that the initial wavefunction is $\Psi(\mathbf{x}_i, t_i) = \delta(\mathbf{x} - \mathbf{x}_i)$), then the probability amplitude of detecting a particle at the spacetime point $(\mathbf{x}_f, t_f)$ on the detection screen is the wavefunction

$$\Psi(\mathbf{x}_f, t_f) = K(\mathbf{x}_f, t_f : \mathbf{x}_i, t_i).$$

The probability distribution is then $\rho(\mathbf{x}_f, t_f) = |\Psi(\mathbf{x}_f, t_f)|^2$. Using the approximation (2) and the fact that, since that any complex number can be factorized into a magnitude and phase $K_m = A_m e^{i\phi_m}$, the probability-density-distribution at the detection screen is

$$\rho(\mathbf{x}_f, t_f) = A_0^2 + A_1^2 + 2A_0A_1\cos\left(\Delta\phi - \frac{q\Phi}{\hbar c}\right),$$

where $\Delta\phi = \phi_1 - \phi_0$ is the "mechanical interference" of the waves. If the two amplitudes of the paths above and below the solenoid have equal probability amplitudes, $A_0 = A_1 = A$, then the probability-amplitude distribution on the detection screen is affected by the magnetic flux through the solenoid

$$\rho(\mathbf{x}_f, t_f) = 4A^2\cos^2\left[\frac{1}{2}\left(\Delta\phi - \frac{q\Phi}{\hbar c}\right)\right]. \tag{3}$$

A Phase-space path integral

We now derive the phase-space path integral representation of the propagator of one-dimensional Hamiltonians of the form $H = H(x, p, t) = T(p, t) + V(x, t)$, along the lines of [6]. Notice that we are treating the general case where the Hamiltonian is time dependent and we are not assuming that H's at different times commute, but we are assuming that the Hamiltonian is of the form $T(p, t) + V(x, t)$. The propagator for a Hamiltonian H is

$$K = \langle x_f | \, U(t_f, t_i) \, | x_i \rangle \, .$$

Feynman's idea [3] was to use the transitive property of time-evolution operators and split the total time of time-evolution, $t_f - t_i$, into $N + 1$ smaller intervals $\delta t = (t_f - t_i)/(N + 1)$, so that the propagator is

$$K = \langle x_f | \, U(t_f, t_N) U(t_N, t_{N-1}) \cdots U(t_n, t_{n-1}) \cdots U(t_1, t_i) \, | x_i \rangle \, . \tag{4}$$

Next, we can use the completeness of the position basis, to express the propagator as a product integrals of propagators across smaller intervals of time δt,

$$K = \prod_{n=1}^{N} \left[\int_{-\infty}^{\infty} dx_n \right] \prod_{n=1}^{N+1} \langle x_n | \, U(t_n, t_{n-1}) \, | x_{n-1} \rangle \, .$$

Now, we to show that the propagator is a sum over all possible paths, we must take the limit as the small time interval δt becomes infinitesimal, or alternatively the number of slices $N + 1$ goes to infinity. Mathematically, this amounts to writing the propagator as the limit

$$K = \lim_{N \to \infty} K_N ,$$

where the truncated propagators K_N are

$$K_N = \prod_{n=1}^{N} \left[\int_{-\infty}^{\infty} dx_n \right] \prod_{n=1}^{N+1} \langle x_n | \, e^{=i \, \delta t \, H(t_n)/\hbar} \, | x_{n-1} \rangle \, .$$

Notice, that since in the limit that $N \to \infty$ the time intervals become infinitesimal, $\delta t \to 0$, we can approximate the time evolution operator across an interval δt by

$$U(t_n, t_{n-1}) = \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_{n-1}}^{t_n} dt H(t) \right] \approx \exp \left[-\frac{i \delta t}{\hbar} H(t_n) \right] .$$

Now, lets focus on the infinitesimal-time propagators $\langle x_n | \, e^{-i \, \delta t \hat{H}(t_n) . \hbar} \, | x_{n-1} \rangle$. Using the *Baker-Campbell-Hausdorf* formula, as discussed in [6], we can separate the Hamiltonian in the exponent into a product of two exponentials, insert a complete set of position eigenstates and a complete set of momentum eigenstates so that $\hat{x}$ and $\hat{p}$ act multiplicatively, we obtain

$$\langle x_n | \, e^{-i \, \delta t \hat{H}(t_n) . \hbar} \, | x_{n-1} \rangle \approx \int_{-\infty}^{\infty} \frac{dp_n}{2\pi\hbar} \exp \left[\frac{i}{\hbar} \left(p_n \frac{(x_n - x_{n-1})}{\delta t} - H(x_n, p_n, t_n) \right) \delta t \right] .$$

Using this re-writing in the expression for K_N, we can write the propagator K as

$$K = \lim_{N \to \infty} \prod_{n=1}^{N} \left[\int_{-\infty}^{\infty} dx_n \right] \prod_{n=1}^{N+1} \left[\int_{-\infty}^{\infty} \frac{dp_n}{2\pi\hbar} \right] \exp \left[\frac{i}{\hbar} \sum_{n=1}^{N+1} \delta t \left(p_n \frac{(x_n - x_{n-1})}{\delta t} - H(x_n, p_n, t_n) \right) \right] .$$

In the limit that $N \to \infty$, the sum in the exponent becomes an integral over time of the classical Lagrangian $L = p\dot{x} - H$, and the the propagator can be written as

$$K = \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x \int \mathcal{D}p \, e^{\frac{i}{\hbar} S[x,p]} ,$$

where $S[x, p]$ is the classical action of the particle.

B Homotopy Theory

Topology is the mathematical study of topological invariants of a space, i.e., those properties of a space which are preserved under continuous deformations of that space. There are many topological invariants which one could study, but as we discuss throughout these notes, holes are especially important for physics and interesting in their right. Thus, we briefly review homotopy theory - the study of holes of a space - in this appendix. A detailed discussion of homotopy theory can be found in Algebraic Topology by Hatcher [5].

Lets begin with some terminology and definitions used in topology. A topological space is a space which has the structure of a topology, i.e., a topology gives a space the notion of relations between points of that space. So, two points p and p' on a topological space can be said to be nearer than the points p and p'', for instance. This in turn gives the space features such as holes and an orientation which are preserved under continuous deformations of the space. More specifically and generally, a continuous deformation between spaces $\mathcal{M}$ and $\mathcal{N}$, called a **homemorphism**, is a continuous function $f : \mathcal{M} \to \mathcal{N}$ such that its inverse $f^{-1} : \mathcal{N} \to \mathcal{M}$ is also continuous.

The question that homotopy theory answers is how do we characterize holes in a space. For that matter, how can a hole be even defined? The answer is given by what is called a **homotopy**. Let $\gamma : [0, 1] \to \mathcal{M}$ be a path in a topological space $\mathcal{M}$. If $\gamma(0) = \gamma(1) = p \in \mathcal{M}$, then γ is a loop based at p. A loop can also be thought as a homeomorphism from the circle to the space $\mathcal{M}$, $\gamma : S^1 \to \mathcal{M}$. A homotopy is a continuous interpolation between two loops. A homotopy is a continuous deformation of a loop to some other loop (this could be defined for paths as well, but we are only interested in loops in these notes). Mathematically, a homotopy between the loops $\gamma_0 : [0, 1] \to \mathcal{M}$ and $\delta : [0, 1] \to \mathcal{M}$, is a continuous one-parameter family of loops (or paths),

$$H_s : [0, 1] \to \mathcal{M}, \qquad s \in [0, 1],$$

such that $H_0(t) = \gamma(t)$ and $H_1(t) = \delta(t)$ for all $t \in [0, 1]$. If a homotopy between two loops exists, then we say that γ is homotopic to δ. It is easy and instructive to show that the relation "homotopic to" is an equivalence relation. Then, the space of loops based at p can be partitioned into equivalence classes of loops $[\gamma]$ under the equivalence relation of homotopy. We can use this equivalence relation to define a group: the **fundamental group** $\pi_1(\mathcal{M})$. The fundamental group, or first homotopy group, is defined to be the quotient space

$$\pi_1(\mathcal{M}, p) = \{\text{loops based at } p\} \Big/ _{\text{homotopy}},$$

where the group multiplication is given by *concatenation* of (equivalence classes of) loops . [The concatenation of two loops γ and δ (which is easily extended to concatenation of paths), such that $\gamma(1) = \delta(0)$, is defined to be

$$\gamma \cdot \delta(t) = \begin{cases} \gamma(2t) & 0 \le t \le \frac{1}{2} \\ \delta(2t-1) & \frac{1}{2} \le t \le 1 \end{cases},$$

where γ and δ must be traversed twice as fast so that $\gamma \cdot \delta$ is traversed in unit time. The group multiplcation for $\pi_1(\mathcal{M}, p)$ is then given by the product of classes $[\gamma] \cdot [\delta] = [\gamma \cdot \delta]$. It is easy to show that this is indeed a group multiplication.] It is also very instructive to show that $\pi_1(\mathcal{M}, p)$ is a group. If the space $\mathcal{M}$ is path connected, then it can be easily shown that $\pi_1(\mathcal{M}, p) = \pi_1(\mathcal{M}, q)$, i.e., the fundamental group is independent of the base point of loops, and so we write $\pi_1(\mathcal{M})$.

As the name *first homotopy group* suggests, there are higher order homotopy groups, defined by higher-dimensional notions of homotopy. Higher dimensional "n-loops" f are homeomorphisms from the n-sphere to the space $\mathcal{M}$

$$f : S^n \to \mathcal{M}.$$

Homotopy between such "higher-dimensional loops" can be defined to provide a way of partitioning the space of "higher-dimensional loops" into equivalence classes, and thus defining the group $\pi_n(\mathcal{M})$ with group multiplication analogous to the one-dimensional case.

The idea behind the use of a homotopy is that, if we can homotopically deform a loop γ based at p into the trivial loop $\delta = \mathbb{I}_{\mathcal{M}}$ based at p - i.e., for all $t \in [0, 1]$, $\delta(t) = p$ - then what we have done is shrunken the loop γ down to a point, p. This implies that there were no obstructions in the way of the homotopy, since homotopy is continuous. If we could not deform a loop to a point, then we say that there is a hole in the space.

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